

PHANENDRA PALADUGU

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SUMMARY

AI Engineer and Python developer with 5+ years building data and ML systems and hands-on experience designing production LLM and agentic systems that reason, plan, and act. Build multi-step agents with LangGraph and LangChain using tool use and function calling, RAG over vector databases, and prompt optimization (chain-of-thought, few-shot, structured outputs) across frontier models (Claude, GPT-4o) and open-source LLMs. Shipped an agentic router that directs queries between text-to-SQL, document RAG, and human-approved action tools, with token, latency, and trace observability. Fine-tuned a 7B model with QLoRA and GRPO that outperformed GPT-4o on BIRD-PG.

TECHNICAL SKILLS

Languages & Core: Python (async/asyncio), SQL, Bash, FastAPI, Docker, Kubernetes (KEDA), CI/CD, GitHub Actions

Agentic AI: LangGraph, LangChain, Tool Use / Function Calling, ReAct, Plan-and-Execute, Human-in-the-Loop, Multi-Agent Routing, MCP (Model Context Protocol)

LLMs & APIs: Claude (Anthropic), GPT-4o (OpenAI), DeepSeek, Llama 3.1, Qwen2.5-Coder, Ollama, LiteLLM, Hugging Face

RAG & Retrieval: RAG, Vector Databases (pgvector, ChromaDB, FAISS), BGE-M3, Hybrid Retrieval, Reranking, Chunking Strategies, Citations

Prompt Engineering: Chain-of-Thought, Few-Shot, Structured Outputs, System-Prompt Optimization, Guardrails, Hallucination Detection

Fine-Tuning: QLoRA, GRPO (RL-based), GGUF Quantization, HuggingFace Hub

LLMOps & Eval: Langfuse (token usage, latency, tracing), Prometheus/Grafana, Evidently AI, Agent/RAG Evaluation (retrieval metrics, LLM-judge faithfulness), MLflow, JWT Auth, pytest

Cloud & ML Foundation: AWS (S3, SageMaker, EC2), GCP (BigQuery, Vertex AI), Azure; PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, PySpark, SpaCy, NER, OCR (Tesseract/DocTR)

EXPERIENCE

AI Engineer | Wells Fargo (Indiana)

Jan 2025 - Present

- Build RAG-based document Q&A and assistant systems using LangChain, ChromaDB, and Sentence Transformers for natural-language search over internal policy documents, compliance manuals, and regulatory filings.
- Develop multi-step LLM pipelines for document classification, entity extraction, and summarization of financial reports using SpaCy, Hugging Face Transformers, and fine-tuned and frontier LLMs.
- Deploy LLM applications as JWT-secured, async FastAPI endpoints with prompt guardrails, hallucination detection, and citation verification for regulatory compliance.
- Apply prompt-engineering techniques (few-shot, chain-of-thought, structured outputs, system-prompt optimization) and design chunking strategies (paragraph, sliding-window, semantic) to raise retrieval quality and reduce hallucination.
- Build embedding and indexing pipelines with Sentence Transformers and ChromaDB/FAISS over thousands of internal documents, with tool-augmented retrieval feeding downstream agents.
- Build evaluation and monitoring frameworks tracking retrieval quality (Recall@K, MRR), answer accuracy, token usage, and latency to benchmark and improve deployed LLM systems.

Lab Assistant | Purdue University (Indiana)

Aug 2024 - Dec 2024

- Coached students through electrical measurements and circuit debugging, and reviewed lab notebooks and technical reports for data accuracy and analytical reasoning.

Technology Analyst | Infosys (India)

Jan 2020 - Dec 2021

- Designed and deployed an NLP deep learning pipeline in TensorFlow/Keras to classify 200+ monthly incidents, extract intent, and auto-recommend resolution articles via text embeddings and cosine similarity, cutting manual triage by 30%.
- Built a knowledge-article retrieval system using TF-IDF and semantic text matching to surface relevant documentation, a precursor to modern RAG workflows.

- Developed text-preprocessing and feature-extraction pipelines using NLTK, SpaCy, and Pandas for tokenization, named entity recognition, and embedding generation on unstructured text.
- Integrated NLP models into Flask REST APIs for real-time inference with prompt templating, response logging, and versioned model endpoints.

Senior Systems Engineer | Infosys (India)

Jan 2017 - Dec 2019

- Built Python automation and ETL with API integrations across SAP BW/APO, processing large-scale enterprise datasets and saving 800+ hours annually.
- Developed a Python email-monitoring application with rule-based NLP to auto-classify and route requests, generating ServiceNow tickets for 24/7 automated processing.
- Wrote complex SQL for data extraction and validation across SAP BW, supporting ad-hoc reporting for global supply chain operations.

PROJECTS

OmniAsk: Production Agentic AI Assistant | [GitHub](#)

Python (async), LangGraph, LiteLLM, Claude, GPT-4o, DeepSeek, FastAPI, PostgreSQL/pgvector, Ollama, BGE-M3, MCP, Docker, Kubernetes (KEDA), Prometheus/Grafana, Langfuse

- Built a production agentic system where a LangGraph router reasons over each request and routes it to the right tool: read-only text-to-SQL, document RAG, or a human-approved action (human-in-the-loop).
- Implemented tool use and function calling across a LiteLLM gateway spanning frontier models (Claude, GPT-4o) and open-source models (DeepSeek, fine-tuned Slonik on Ollama), with database-level guardrails for safe execution.
- Built RAG with chunk-level citations using BGE-M3 hybrid retrieval and reranking, and exposed agent state and tools through MCP (Model Context Protocol).
- Served the system over a streaming async FastAPI API and web UI with Postgres-backed checkpointing for multi-step agent runs.
- Instrumented full observability with Langfuse and Prometheus/Grafana, tracking token usage, latency, and failure modes, and deployed on Kubernetes with KEDA autoscaling.

Slonik-7B-GRPO: Text-to-SQL Fine-Tuned LLM | [GitHub](#) | [HuggingFace](#)

Python, PyTorch, QLoRA, GRPO, Qwen2.5-Coder-7B, BIRD-SQL, PostgreSQL, SQLite, GGUF, Ollama

- Fine-tuned Qwen2.5-Coder-7B with QLoRA and GRPO (execution-based reinforcement learning) for PostgreSQL and SQLite text-to-SQL, with database-level guardrails for read-only query generation.
- Raised BIRD-PG accuracy from 12.22% to 38.20%, outperforming GPT-4o (34.44%) at 7B parameters, and published GGUF quantizations for local deployment via Ollama and llama.cpp.

DocBrain: Enterprise Document Intelligence Engine | [GitHub](#)

Python, ChromaDB, Sentence-Transformers, Llama 3.1 8B, Tesseract/DocTR (OCR), SpaCy, FastAPI (JWT), Docker

- Built a grounded RAG pipeline over 4 document types with quality-based OCR routing and SpaCy NER; evaluated on 30 queries (0.90 Recall@5, 0.78 MRR, 1.00 Recall@1 on contracts) plus an LLM-judge faithfulness score, secured behind JWT auth, with 162 pytest tests.

EDUCATION

Master of Science, Electrical and Computer Engineering

GPA 3.61/4

Purdue University Northwest, Hammond, Indiana

2025

Coursework: Machine Learning, Neural Networks, Big Data, Computer Vision, Digital Signal Processing, Computer Networks